



Oracle Database Administration

I

Chapter 6: Managing Control Files, Online Redo Logs, and Archive logs

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Objectives:

1. Multiplex control file
2. Add new control file
3. Multiplex Redo log files
4. Understand log switch operations
5. Manage redo log files
6. Implement the database in Archive mode

1. Managing Control Files:

We are going to discuss typical control file maintenance tasks, such as adding, moving, and removing control files.

A control file is a small binary file that describes the structure of the database and it is created when the DB is created. It must be available for writing by the Oracle server whenever the database is mounted or opened.

Without this file, the database cannot be mounted, and recovery or re-creation of the control file is required. Your database should have a minimum of two control files on different storage devices to minimize the impact of a loss of one control file.

The loss of a single control file causes the instance to fail because all control files must be available at all times. However, recovery can be a simple matter of copying one of the other control files. The loss of all control files is slightly more difficult to recover from but is not usually catastrophic.

1.1. Control File Contents:

A control file contains the following entries:

- Database name
- Names and locations of data files
- Names and locations of online redo log files
- Current online redo log sequence number
- Checkpoint information
- Names and locations of RMAN backup files

1.2. Getting Information about Control File:

Every Oracle database must have at least one control file. When you start your database in nomount mode, the instance is aware of the location of the control files from the `CONTROL_FILES` initialization parameter in the `spfile` or `init.ora` file. When you issue a `STARTUP NOMOUNT` command, Oracle reads the parameter file and starts the background processes and allocates memory structures:

```
-- locations of control files are known to the instance
SQL> startup nomount;
```

At this point, the control files have not been touched by any processes. When you alter your database into mount mode, the control files are read and opened for use:

```
-- control files opened
SQL> alter database mount;
```

If any of the control files listed in the `CONTROL_FILES` initialization parameter are not available, then you cannot mount your database.

When you successfully mount your database, the instance is aware of the locations of the data files and online redo logs but has not yet opened them.

After the database has been opened, Oracle will frequently write information to the control files, such as when you make any physical modifications (e.g., creating a tablespace, adding/removing/resizing a data file). Oracle writes to all control files specified by the `CONTROL_FILES` initialization parameter. If Oracle cannot write to one of the control files, an error is thrown:

```
ORA-00210: cannot open the specified control file
```

You can query much of the information stored in the control file from data dictionary views.

This example displays the types of information stored in the control file by querying `v$controlfile_record_section`:

```
SQL> select distinct type from v$controlfile_record_section;
```

Here is a partial listing of the output:

```
TYPE
-----
FILENAME
TABLESPACE
RMAN CONFIGURATION
BACKUP CORRUPTION
PROXY COPY
```


FLASHBACK LOG
REMOVABLE RECOVERY FILES
AUXILIARY DATAFILE COPY
DATAFILE

If your database is in a nomount state, a mounted state, or an open state, you can view the names and locations of the control files, as follows:

```
SQL> show parameter control_files
```

You can also view control file location and name information by querying the V\$CONTROLFILE view. This query works while your database is mounted or open:

```
SQL> select name from v$controlfile;
```

1.3. Multiplexing the Control File:

Since control file is critical for the database operation, Oracle recommends a minimum of two control files. Multiplexing is defined as keeping a copy of the same control file in different locations

1.4. Adding a Control File:

The basic procedure for adding a control file is as follows:

1. Alter the initialization file CONTROL_FILES parameter to include the new location and name of the control file.
2. Shut down your database.
3. Use an OS command to copy an existing control file to the new location and name.
4. Restart your database.

Depending on whether you use a spfile or an init.ora file, the previous steps vary slightly:

Spfile Scenario

If your database is open, you can quickly determine whether you are using a spfile with the following SQL statement:

```
SQL> show parameter spfile
```

When you have determined that you are using a spfile, use the following steps to add a control file:

1. Determine the CONTROL_FILES parameter's current value:

```
SQL> show parameter control_files
```

2. Alter your CONTROL_FILES parameter to include the new control file that you want to add, but limit the scope of the operation to the spfile (you cannot modify this parameter in memory). Make sure you also include any control files listed in step A:

```
SQL> alter system set control_files  
='/u01/dbfile/o18c/control01.  
ctl','/u01/dbfile/o18c/control02.ctl' scope=spfile;
```

3. Shut down your database:

```
SQL> shutdown immediate;
```

Copy an existing control file to the new location and name. In this example, a new control file named control02.ctl is created via the OS cp command:

```
$ cp /u01/dbfile/o18c/control01.ctl /u01/dbfile/o18c/  
control02.ctl
```

4. Start up your database:

```
SQL> startup;
```

You can verify that the new control file is being used by displaying the CONTROL_FILES parameter:

```
SQL> show parameter control_files
```

Init.ora Scenario

Run the following statement to verify that you are using an init.ora file. If you are not using a spfile, the VALUE column is blank:


```
SQL> show parameter spfile
NAME                                TYPE                                VALUE
-----                                -
```

To add a control file when using a text `init.ora` file, perform the following steps:

1. Shut down your database:
SQL> shutdown immediate;
2. Edit your `init.ora` file with an OS utility (such as `vi`), and add the new control file location and name to the `CONTROL_FILES` parameter. This example opens the `init.ora` file, using `vi`, and adds `control02.ctl` to the `CONTROL_FILES` parameter:

```
$ vi $ORACLE_HOME/dbs/inito19c.ora
```

Listed next is the `CONTROL_FILES` parameter after `control02.ctl` is added:

```
control_files='/u01/dbfile/o18c/control01.ctl',
'/u01/dbfile/o18c /control02.ctl'
```

3. From the OS, copy the existing control file to the location and name of the control file being added:

```
$ cp /u01/dbfile/o18c/control01.ctl
/u01/dbfile/o18c/control02.ctl
```

4. Start up your database:

```
SQL> startup;
```

You can view the control files in use by displaying the `CONTROL_FILES` parameter:

```
SQL> show parameter control_files
```


2. Online Redo Logs:

We are going to examine DBA activities related to online redo log files, such as renaming, adding, dropping, and relocating these critical files. Online redo logs store a record of transactions that have occurred in your database.

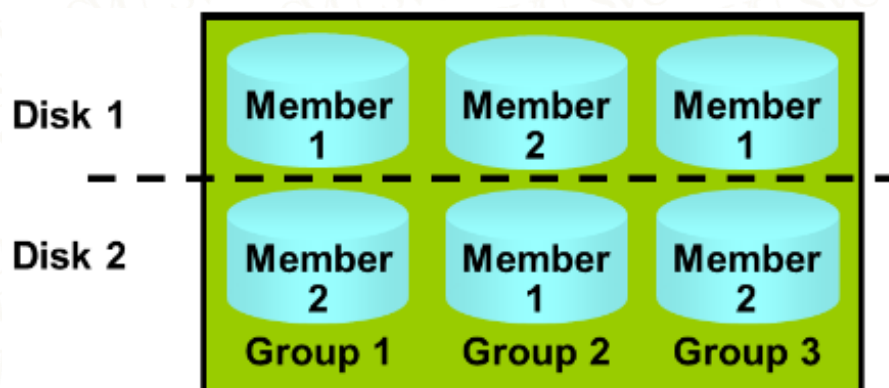
These logs serve the following purposes:

- Provide a mechanism for recording changes to the database so that in the event of a media failure, you have a method of recovering transactions
- Ensure that in the event of total instance failure, committed transactions can be recovered (crash recovery) even if committed data changes have not yet been written to the data files
- Allow administrators to inspect historical database transactions through the Oracle LogMiner utility
- They are read by Oracle tools such as GoldenGate or Streams to replicate data

2.1. Multiplexing the Redo Log:

You are required to have at least two online redo log groups in your database. Each online redo log group must contain at least one online redo log member. The member is the physical file that exists on disk. You can create multiple members in each redo log group, which is known as multiplexing your online redo log group.

It is highly recommended that you multiplex your online redo log groups and, if possible, have each member on a separate physical device governed by a separate controller.



Multiplexing redo logs may impact overall database performance as it increases database I/O.

The log-writer process flashes log buffer (in the SGA) to the online redo log files (on disk). The redo record has a system change number (**SCN**) assigned to it in order to identify the transaction redo information. There are committed and uncommitted records written to the redo logs. The log writer flushes the contents of the redo log buffer when any of the following are true:

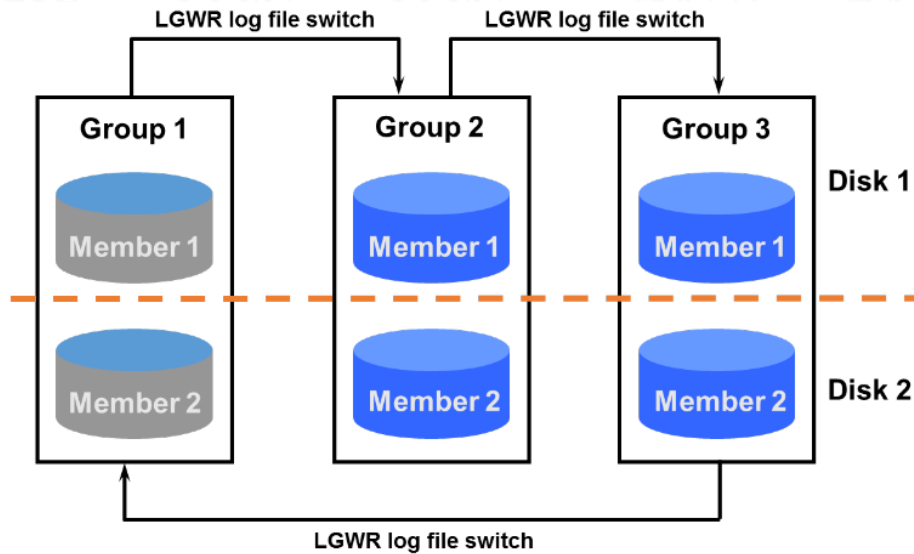
- A COMMIT is issued
- A log switch occurs
- Three seconds go by
- The redo log buffer is one-third full.

Since this is a database process, the container database (**CDB**) will manage the redo logs. **PDBs** do not have their own redo logs, which also means that planning for space and sizing of the redo logs is at the CDB level and includes all of the PDB transactions.

2.2. Log Switch Description:

The online redo log group that the log writer is actively writing to is the current online redo log group. The log writer writes simultaneously to all members of a redo log group. The log writer needs to successfully write to only one member in order for the database to continue operating. The database ceases operating if the log writer cannot write successfully to at least one member of the current group.

When the current online redo log group fills up, a log switch occurs, and the log writer starts writing to the next online redo log group. A log sequence number is assigned to each redo log when a switch occurs to be used for archiving. The log writer writes to the online redo log groups in a round-robin fashion:



Because you have a finite number of online redo log groups, eventually the contents of each online redo log group are overwritten. If you want to save a history of the transaction information, you must place your database in archivelog mode.

When your database is in archivelog mode, after every log switch the archiver background process copies the contents of the online redo log file to an archived redo log file. In the event of a failure, the archived redo log files allow you to restore the complete history of transactions that have occurred since your last database backup.

2.3. Mechanisms log files to protect your database:

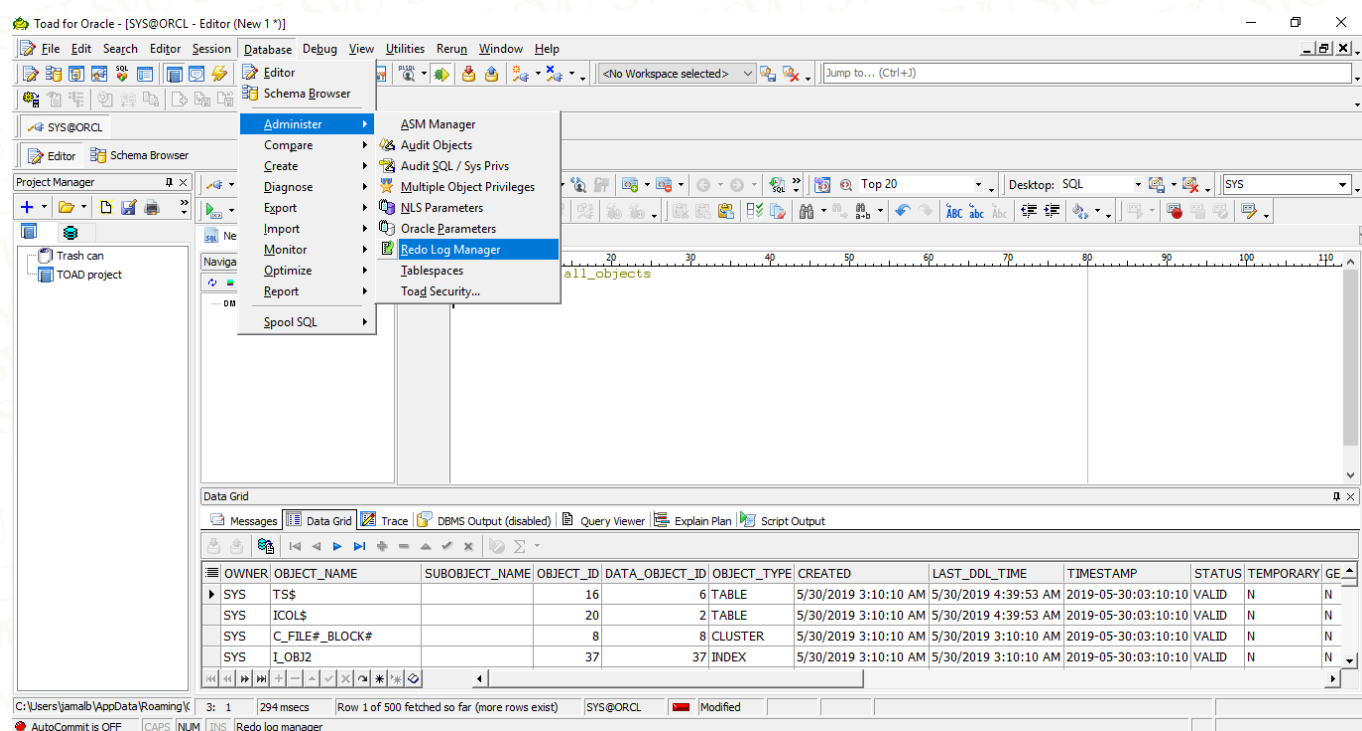
The contents of the current online redo log files are not archived until a log switch occurs. This means that if you lose all members of the current online redo log file, you lose transactions. Listed next are several mechanisms log files:

- Multiplex the groups
- Consider setting the `ARCHIVE_LAG_TARGET` initialization parameter to ensure that the online redo logs are switched at regular intervals
- If possible, never allow two members of the same group to share the same physical disk
- Ensure that OS file permissions are set appropriately (restrictive, that only the owner of the Oracle binaries has permissions to write and read).

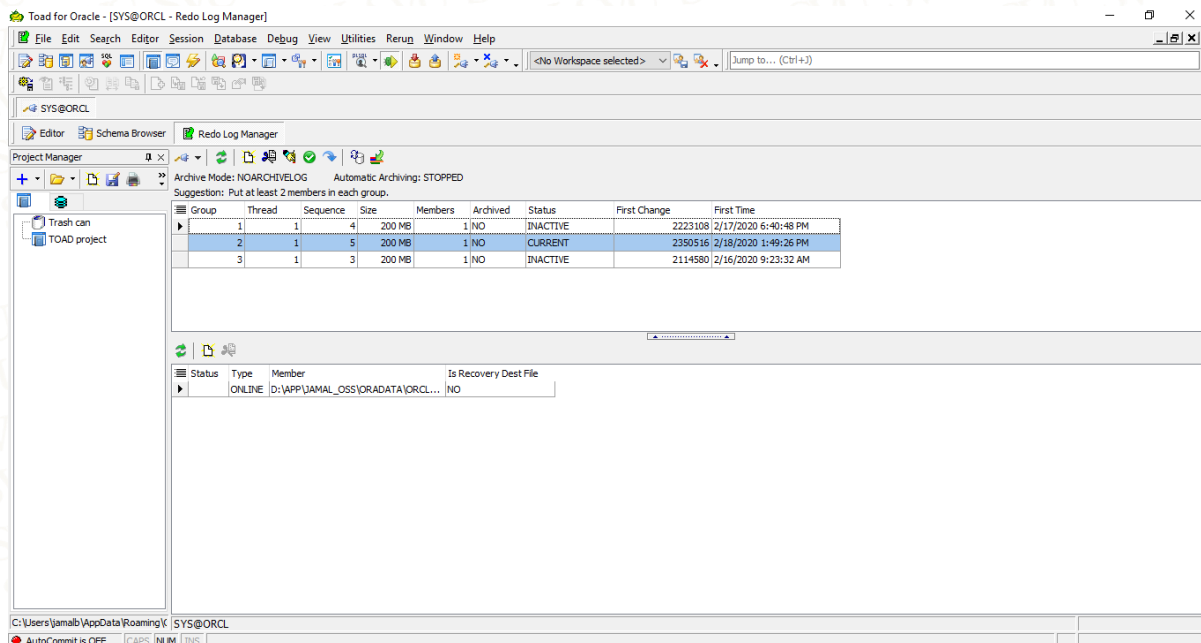
- Use physical storage devices that are redundant (i.e., RAID [redundant array of inexpensive disks])
- Appropriately size the log files, so that they switch and are archived at regular intervals

Please click on the button to see how to choose and monitor Redolog groups from Toad for Oracle

The following figure shows how to choose monitoring Redolog groups from Toad for Oracle:



The following figure shows how to monitor Redolog groups from Toad for Oracle:



2.4. Adding Online Redo Log Groups:

```
ALTER DATABASE ADD LOGFILE GROUP 3
('C:\ORACLE\ORADATA\u01\redo0301.log',
'C:\ORACLE\ORADATA\u01\redo0302.log')
SIZE 1M;
```



You specify the name and location of the members with the file specification. The value of the GROUP parameter can be selected for each redo log file group. If you omit this parameter, the Oracle server generates its value automatically.

2.5. Adding Online Redo Log Members:

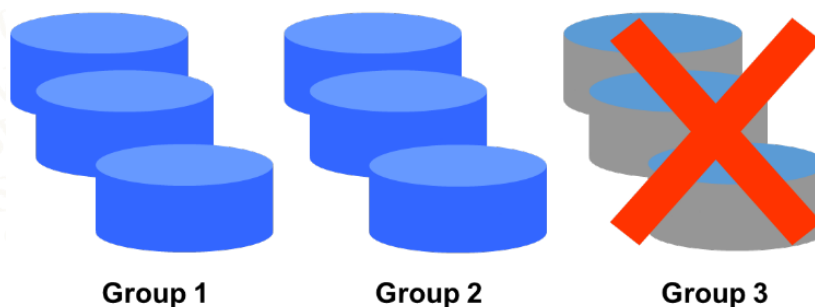
```
ALTER DATABASE ADD LOGFILE MEMBER
'C:\ORACLE\ORADATA\u01\redo0103.log' TO GROUP 1,
'C:\ORACLE\ORADATA\u01\redo0203.log' TO GROUP 2,
'C:\ORACLE\ORADATA\u01\redo0303.log' TO GROUP 3;
```



Use the fully specified name of the log file members; otherwise the files are created in a default directory of the database server. If the file already exists, it must have the same size, and you must specify the REUSE option. You can identify the target group either by specifying one or more members of the group or by specifying the group number.

2.6. Dropping Online Redo Log Groups:

```
ALTER DATABASE DROP LOGFILE GROUP 3;
```



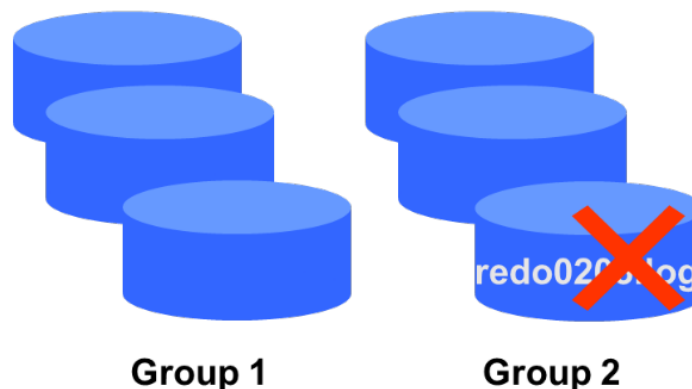
To increase or decrease the size of online redo log groups, add new online redo log groups (with the new size) and then drop the old ones.

Restrictions

- An instance requires at least two groups of online redo log files
- An active or current group cannot be dropped
- When an online redo log group is dropped, the operating system files are not deleted

2.7. Dropping Online Redo Log Members:

```
ALTER DATABASE DROP LOGFILE MEMBER  
'C:\ORACLE\ORADATA\u01\redo0203.log';
```



Restrictions

- If the member you want to drop is the last valid member of the group, you cannot drop that member
- If the group is current, you must force a log file switch before you can drop the member
- If the database is running in ARCHIVELOG mode and the log file group to which the member belongs is not archived, then the member cannot be dropped
- When an online redo log member is dropped, the operating system file is not deleted

2.8. Obtaining Log files Information:

Use the `v$log` and `v$logfile` views to display information about online redo log groups and corresponding members:

```
SELECT a.group# ,a.thread#,a.status grp_status,b.member member
,b.status mem_status,a.bytes/1024/1024 mbytes
FROM v$log a, v$logfile b
WHERE a.group# = b.group#
ORDER BY a.group#, b.member;
```

Status for Online Redo Log Groups in the `v$log` View:

Status	Meaning
CURRENT	The log group is currently being written to by the log writer.
ACTIVE	The log group is required for crash recovery and may or may not have been
CLEARING	The log group is being cleared out by an ALTER DATABASE CLEAR
CLEARING_CURRENT	The current log group is being cleared of a closed thread.
INACTIVE	The log group is not required for crash recovery and may or may not have
UNUSED	The log group has never been written to; it was recently created.

Status for Online Redo Log File Members in the `v$logfile` View:

Status	Meaning
INVALID	The log file member is inaccessible or has been recently created.
DELETED	The log file member is no longer in use.
STALE	The log file member's contents are not complete.
NULL	The log file member is being used by the database.

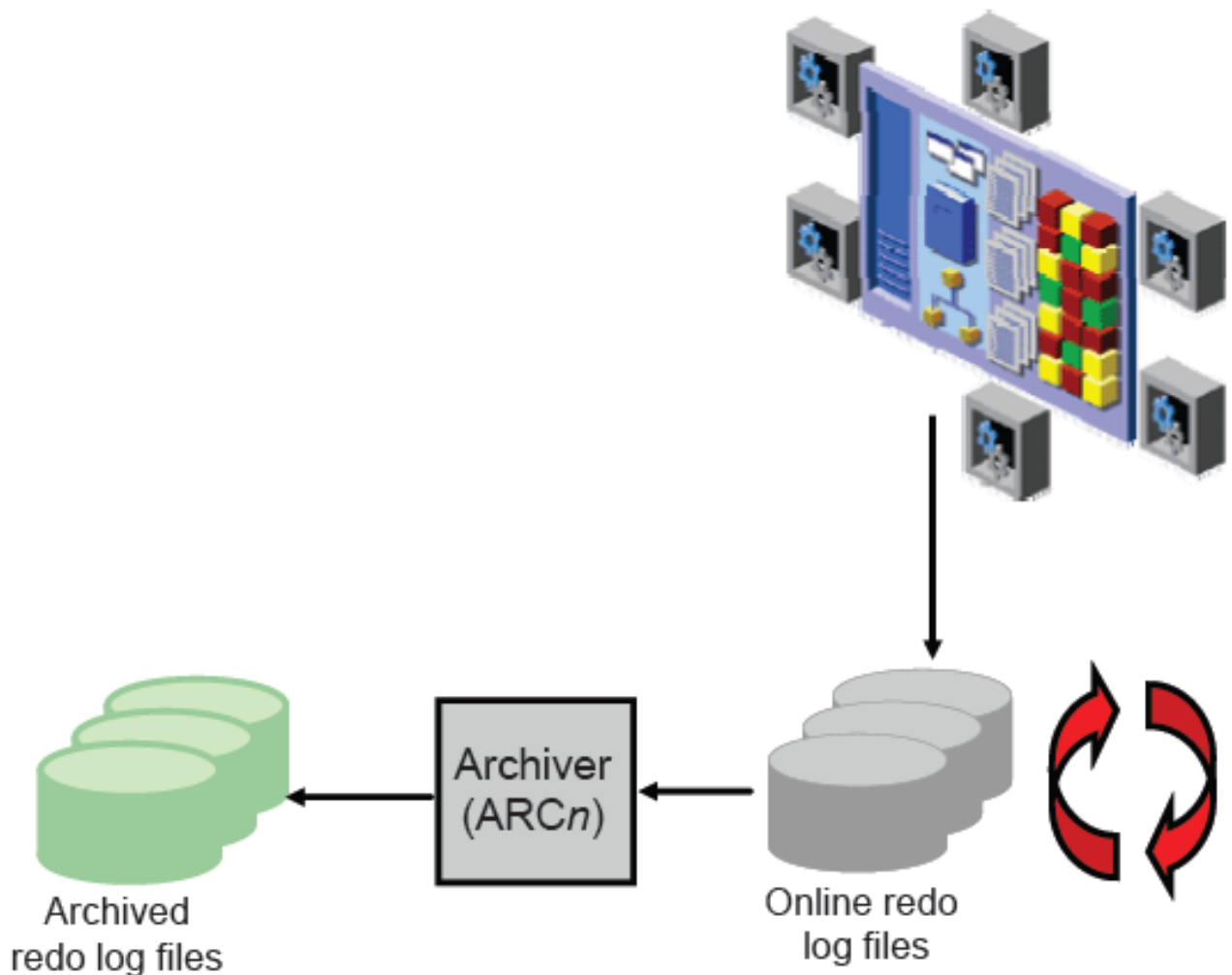
3. Implementing Archivelog Mode:

We are going to ensure that the architectural aspects of enabling and implementing archiving are covered.

Creating Archived Redo Log Files

To preserve redo information and configure your database for maximum recoverability, create archived copies of redo log files by performing the following steps:

- Specify archived redo log file-naming convention
- Specify one or more archived redo log file locations
- Place the database in ARCHIVELOG mode



Archived Redo Log Files

Archivelog mode preserves redo data for the long term by employing an archiver background process to copy the contents of a filled online redo log to what is termed an archive redo log file. The trail of archive redo log files is crucial to your ability to recover the database with all changes intact, right up to the precise point of failure.

The process of switching from one online redo log group to another is called a log switch. The ARCn process initiates archiving of the filled log group at every log switch. It automatically archives the online redo log group before the log group can be reused so that all the changes made to the database are preserved. This enables recovery of the database to the point of failure even if a disk drive is damaged.

One of the important decisions that a DBA must make is whether to configure the database to operate in ARCHIVELOG mode or in NOARCHIVELOG mode.

- In NOARCHIVELOG mode, the online redo log files are overwritten each time a log switch occurs. This is the default mode of any created database. recovery is possible only until the time of the last backup. All transactions made after that backup are lost
- In ARCHIVELOG mode, inactive groups of filled online redo log files must be archived before they can be used again. recovery is possible until the time of the last commit. Most production databases are operated in ARCHIVELOG mode

Note: Back up your database after switching to ARCHIVELOG mode because your database is recoverable only from the first backup taken in that mode.

Information about archived logs can be obtained from V\$INSTANCE.

```
SQL> SELECT archiver FROM v$instance;
```

Some Archiving Parameters

LOG_ARCHIVE_DEST: Destination to write archive log files, it must be a valid directory in the server where the database is located

LOG_ARCHIVE_START: Whether oracle should enable automatic archiving

Note: To set the value of the previous parameters, use alter system set command.

Enable ARCHIVELOG Mode

- Shutdown the database
- Set up the appropriate parameters
- STARTUP MOUNT
- ALTER DATABASE ARCHIVELOG
- ALTER DATABASE OPEN

Disable ARCHIVELOG Mode

- Shutdown the database
- STARTUP MOUNT
- ALTER DATABASE NOARCHIVELOG
- ALTER DATABASE OPEN

4. Quiz:

1. When the control file is read?
 - a) Nomount stage
 - b) Open stage
 - c) Mount stage
 - d) None of the other options

2. Which of these operations cannot be accomplished while the database is open?
(Choose all correct answers.)
 - a) Adding a controlfile copy
 - b) Adding an online log file member
 - c) Changing the location of the flash recovery area
 - d) Changing the archivelog mode of the database

3. There are several steps involved in transitioning to archivelog mode. Put these in the correct order:
 1. alter database archivelog
 2. alter database open
 3. alter system archive log start
 4. full backup
 5. shutdown immediate
 6. startup mount
 - a) 5, 6, 1, 2, 4; 3 not necessary
 - b) 5, 4, 6, 1, 2, 3
 - c) 6, 1, 3, 5, 4, 2
 - d) 1, 5, 4, 6, 2; 3 not necessary
 - e) 5, 6, 1, 2, 3; 4 not necessary

4. A control file is a small binary file that describes the structure of the database and it is created by DBA after the DB is created.
 - a) True
 - b) False
5. A control file must be available for writing by the Oracle server whenever the database is mounted or opened?
 - a) True
 - b) False
6. The loss of a single control file causes:
 - a) The instance to continue starting up
 - b) The instance to continue mounting
 - c) All the other options
 - d) The instance to fail because all control files must be available at all times
7. What is the incorrect statement related to the contents of the control file?
 - a) Database name
 - b) Names and locations of data files
 - c) Names and locations of online redo log files
 - d) Current online redo log sequence number
 - e) None of the other options
8. In any state you can view the names and locations of the control files?
 - a) nomount state
 - b) Mount state
 - c) open state
 - d) all the other options

9. What does Oracle do if it cannot write to one of the control files?
- a) Continue normally
 - b) An error is thrown
 - c) Try to create a new control file
 - d) Try to delete that control file
10. Online Redo logs Provide a mechanism for recording changes to the database so that in the event of a media failure, you have a method of recovering transactions?
- a) True
 - b) False
11. You are required to have at least two online redo log groups in your database?
- a) Three online redo log groups
 - b) One online redo log groups
 - c) Two online redo log groups
 - d) Four online redo log groups
12. Why Multiplexing redo logs may impact overall database performance as it increases database I/O?
- a) As it increases database I/O
 - b) As it decreases database I/O
 - c) Because it causes writing in all data files
 - d) Because it causes writing in all redolog groups
13. The log-writer process flashes log buffer (in the SGA) to the online redo log files (on disk)?
- a) True
 - b) False

14. The log-writer process flashes log buffer (in the SGA) to the online redo log files (on disk)?
- a) To the related data files (on disk)
 - b) To the online redo log files (on disk)
 - c) To the related tablespaces (on disk)
 - d) To the related segments (on disk)
15. When does a log switch occur?
- a) There is no log switch as there is one log group
 - b) When the current data file fills up
 - c) When the current control file fills up
 - d) When the current online redo log group fills up
16. The log writer writes to the online redo log groups in a round-robin fashion?
- a) True
 - b) False
17. What is the name of the view that gives the status for Online Redo Log Groups?
- a) V\$LOGFILE View
 - b) V\$LOG View
 - c) V\$LOGSESSION View
 - d) V\$LOGINSTANCE View
18. What is the name of the view that gives the status for Online Redo Log File Members?
- a) V\$LOGFILE View
 - b) V\$LOG View
 - c) V\$LOGSESSION View
 - d) V\$LOGINSTANCE View

19. What is the step that doesn't belong to create archived copies of redo log files?

- a) Specify archived redo log file-naming convention.
- b) Specify one or more archived redo log file locations.
- c) Place the database in ARCHIVELOG mode.
- d) Place the BACKUP plan for each tablespace

20. What is the statement that doesn't related to ARCHIVELOG mode?

- a) inactive groups of filled online redo log files must be archived before they can be used again
- b) Recovery is possible until the time of the last commit
- c) The online redo log files are overwritten each time a log switch occurs
- d) Most production databases are operated in ARCHIVELOG mode.

Quiz answers – chapter 6	
1	c
2	a, d
3	a
4	b
5	a
6	d
7	e
8	d
9	b
10	a
11	c
12	a
13	a
14	b
15	d
16	a
17	b
18	a
19	d
20	c

References:

Books

1. Oracle Database Administrator's Guide, 18c (Oracle University), March 2019, Primary Author: Rajesh Bhatiya, Randy Urbano
2. database-new-features-guide (Oracle University), 18c E88909-01 February 2018, Primary Authors: Tanaya Bhattacharjee, Sunil Surabhi, Mark Baue
3. The Cloud DBA-Oracle: Managing Oracle Database in the Cloud – First Edition, Abhinivesh Jain and Niraj Mahajan. Apress – ISBN-13 (pbk): 978-1-4842-2634-6 ISBN-13 (electronic): 978-1-4842-2635-3

Web Sites

1. <https://docs.oracle.com/database>
2. <https://docs.oracle.com/en/database/oracle/oracle-database/19/whats-new.html>
3. <http://www.oracle.com/technetwork/database/enterprise-edition/downloads/index.html>